

DORA YANG

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WORK EXPERIENCE

Sibel Health

Systems Engineer II

Systems Engineer I

Chicago, IL, USA

Jun. 2025 - Present

Sep. 2024 - Jun. 2025

- Project lead and sole systems engineer for a next-generation wireless pulse oximeter. Leading a 12-person cross-functional team (mechanical, electrical, firmware, algorithm, and manufacturing) through design reviews, scheduling, design transfer, and verification activities, including design history file management.
- Authored functional (vitals capture, data transmission) and performance (range, accuracy, battery life) system requirements, ensuring traceability and coverage across ~80 requirements and ~150 test cases.
- Built and maintained the risk management file by assessing probability and severity of hazards, mapping to and defining risk controls, and assessing residual risk after verification.

Hardware Test Engineer

Jan. 2024 - Sep. 2024

- Performed PCBA bring up and performance testing using bench equipment (power analyzers, oscilloscopes, multimeters, soldering station): power rail measurements, current consumption profiling, firmware flashing, and functional vital signs validation using calibrated simulators.
- Executed environmental verification testing (ingress protection, shipping/transport, thermal safety, and temperature/pressure cycling); coordinated with external test labs for request for proposals, quotes/purchase orders, protocols, on-site supervision and report review.
- Designed and programmed an automated adhesive peel test fixture by sourcing and integrating a linear stage, optical encoders, and microcontroller, then mounting an off-the-shelf robotic arm and writing a Python control application reducing test time by 80% and eliminating manual operator involvement.

D-Wave Systems

Electrical Engineer Co-op

Burnaby, BC, Canada

May 2022 – Aug. 2022

- Redesigned the power supply for 20 quantum computers, replacing a faulty linear supply with a switch-mode alternative; built the full assembly including wiring, connectors, and a 3D-printed enclosure.
- Wrote STM32 firmware in C++ to bridge UART and RS-232 communication protocols between the PSU and control servers, and developed a Python GUI for setting output voltage levels, toggling channels, and logging communication.
- Designed and built a resistive test load jig with power resistors, heat sinks, cooling fans, toggle switches, panel meters, and thermal fuses; reduced power supply build verification time by 30%.
- Developed a Python supply chain monitoring tool using the Octopart/Nexar API to query real-time stock availability for 2,100+ high-risk BOM components, presented findings to the supply chain team.

Precision Nanosystems Inc. (Now part of Cytiva)

Instrumentation/Test Engineer Co-op

Vancouver, BC, Canada

May 2021 – Dec. 2021

- Contributed to the development of a commercial microfluidic lipid nanoparticle formulation instrument (Cytiva NanoAssemblr®), characterizing fluidic system performance using flow, pressure, and temperature sensors to guide pump and materials selection.
- Built test jigs, authored protocols and reports, streamlined data processing and visualization by writing Python and MATLAB scripts for sensor measurement analysis, sampling distributions, and trend plots.

EDUCATION

Bachelors of Applied Science in Engineering Physics

The University of British Columbia

Sep. 2018 - May 2023

Vancouver, BC, Canada

SKILLS

Software

Python, MATLAB, C++, Microsoft Office Suite (Excel, Word, Powerpoint), Jama, Matrix, Claude

Hardware

KiCAD, Altium, Soldering, Circuit Analysis, Oscilloscopes, Microcontrollers, Signal Generators

Certifications

Architecture of Complex Systems (MIT xPro), Models in Engineering (MIT xPro)

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Innovation, Science and Economic Development Canada

Telecommunications Engineer Co-op

Ottawa, ON, Canada

Jan. 2020 - Apr. 2020

- Contributed to the technical regulations for deploying Wi-Fi and 5G devices in Canada's 6GHz band.
- Assessed and analyzed technical papers from telecommunications and RF stakeholders, focusing on data related to interference, device power, and methods for safeguarding incumbent services.
- Created and presented a Powerpoint update on 6GHz band initiatives to the Deputy Minister of ISED.

PROJECTS

Flooded Cave Rescue Communication System

Oct. 2022 - May 2023

- Collaborated with UBC Physics to create a wireless communication device capable of sending text messages at a rate of 50 bits per second with a range of 80 meters in flooded cave conditions.
- Designed the electrical schematics for a power circuit that supplied 200 Vpp at 200 kHz to drive an acoustic transducer and debugged using oscilloscopes, multimeters, probes and function generators.
- Wrote C++ firmware for communications error handling, boosting system performance and reliability.

Electric Motor Dynamometer

Oct. 2021 - May 2023

- Led a team of 3 students designing the hardware and software for a dynamometer to test UBC Supermileage's electric motor and powertrain systems.
- Successfully integrated a torque PID controller, featuring a hysteresis brake, a microcontroller and optical and force sensors for precise system speed monitoring, data collection and feedback control.
- Engineered a user-friendly control PCB, integrating screens and actuators for intuitive operation.

CANBus Vehicle Accessory PCB

Oct. 2020 - Nov. 2021

- Designed the electrical schematics and PCB layout for a board controlling vehicle accessories (horn, wipers), reducing the size and increasing the throughput by 50% and 40% respectively.
- Assembled and tested the PCB by conducting voltage and current measurements, signal integrity checks and functional testing of each component to ensure it met operational standards.
- Implemented a message handling system using CAN Bus communication protocol in C++.

Parking Simulation (ROS, OpenCV, Python)

Oct. 2020 - Nov. 2020

- Used imitation learning to implement a self-driving robot that navigated through a course 2nd fastest out of 20 teams.
- Applied classical computer vision techniques to isolate license plates in the robot's camera feed.
- Trained a convolution neural network to identify license plate characters with 97% accuracy.

UBC Supermileage Design Team

Oct 2020 - May 2023

- Led a team of students designing a dynamometer to test motors. Integrated a microcontroller, hysteresis brake, and sensors for speed monitoring, data collection and feedback control.
- Designed PCB schematics and layout in KiCAD for a board controlling vehicle accessories (horn, wipers), reducing size and increasing throughput by 50% and 40% respectively.
- Assembled and tested the PCB by conducting voltage and current measurements, signal integrity checks and functional testing of each component to ensure it met operational standards.

UBC SAE Supermileage Design Team

Sep. 2020 - May 2023

- Implemented a message handling system to control vehicle accessories (blinkers, horn, wipers) using Controller Area Network (CAN bus) communication protocol.

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- Designed a programmable dynamometer to test fuel cell and electronic motors.
- Selected electrical components, designed schematic and PCB layout documents, assembled, and tested the CAN accessories and Dyno PCBs that will be used for the competition vehicle.

ACTIVITIES

Events Coordinator - UBC Figure Skating Club

Sep. 2022 - May 2023

Social Committee Lead - UBC Women in Engineering

Sep. 2020 - May 2022

Graphic Designer and Photographer - Amnesty International UBC

Sep. 2018 - May 2022

Telecommunications Engineer Co-op

Jan. 2020 - Apr. 2020

Innovation, Science and Economic Development (ISED) Canada

Ottawa, ON, Canada

- Contributed to the technical regulations for deploying Wi-Fi and 5G devices in Canada's 6GHz band.
- Assessed and analyzed technical papers from 100+ telecommunications and RF stakeholders, focusing on data related to interference, device power, and methods for safeguarding incumbent services.
- Created and presented a Powerpoint update on 6GHz band initiatives to the Deputy Minister of ISED.

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Autonomous Recycling Robot

Jul 2020 - Aug 2020

- Constructed a robot capable of finding and picking up 5 soda cans, line following using a PID controller to a bin and dumping the cans into it in under a minute.
- Generated mechanical drawings of the robot using OnShape CAD software and assembled the robot using corrugated plastic, hand tools and calipers.
- Used voltmeters and ammeters to troubleshoot circuits by analyzing signals and voltage outputs.

EDUCATION

Bachelor's of Applied Science in Engineering Physics

Graduated May 2023

The University of British Columbia, GPA: 3.9/4.33

Vancouver, BC, Canada

PROFESSIONAL EXPERIENCE

Electrical Engineer (Co-op)

May 2022 – Aug. 2022

D-Wave Systems

Burnaby, BC, Canada

- Updated the design of faulty power supply units (PSU) for 20 quantum computers with a microcontroller, UART communication, a graphical user interface, and a 3D printed enclosure designed in Solidworks.
- Decreased the verification time of PSU builds by 30% by designing a voltage monitoring test load.
- Built a Python tool to automate stock monitoring for 2100+ supply chain components.

Instrumentation Engineer (Co-op)

May 2021 – Dec. 2021

Precision Nanosystems

Vancouver, BC, Canada

- Worked on a team designing a medicine formulation device for commercial use.
- Developed procedures, built jigs, and performed tests using sensors, hand tools, and data acquisition equipment to evaluate system performance, regulatory compliance and guide component selection.
- Streamlined raw data processing and analysis by writing Python and MATLAB scripts.

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- Maintained comprehensive records of test procedures/test plans and outcomes and delivered findings and design recommendations to project engineers.

Spectrum Management Engineer (Co-op)

Innovation, Science and Economic Development Canada

Jan. 2020 - Apr. 2020

Ottawa, ON, Canada

- Contributed to the technical regulations for deploying Wi-Fi and 5G devices in Canada's 6GHz band.
- Assessed and analyzed technical papers from telecommunications and RF stakeholders, focusing on data related to interference, device power, and methods for safeguarding incumbent services.
- Created and presented a Powerpoint update on 6GHz band initiatives to the Deputy Minister of ISED.

PROJECTS

Flooded Cave Rescue Communication System

Oct. 2022 - May 2023

- Collaborated with UBC Physics to create a robust under-water wireless communication device capable of sending text messages at a rate of 6 characters per second with a range of 100 meters.
- Designed and debugged a power circuit that supplied 200 Vpp at 200 kHz to drive an acoustic transducer.
- Boosted messaging system reliability and performance by 40% by writing error handling firmware in C++.

Electric Motor Dynamometer

Sep. 2022 - May 2023

- Led a team of 3 students designing a dynamometer to test UBC Supermileage's powertrain systems.
- Successfully integrated a torque PID controller, featuring a hysteresis brake, a microcontroller and optical and force sensors for precise system speed monitoring, data collection and feedback control.
- Engineered a user-friendly control PCB, integrating screens and actuators for intuitive operation.

Supermileage Vehicle Accessory PCB

Oct. 2020 - Nov. 2021

- Designed the electrical schematics and PCB layout for a board controlling vehicle accessories (horn, wipers), reducing the size and increasing the throughput by 50% and 40% respectively.
- Assembled and troubleshooted the PCB by conducting voltage and current measurements, signal integrity checks and functional testing of each component to ensure it met operational standards.
- Implemented a message handling system using CANBus communication protocol in C++.

SKILLS

Software

Python, MATLAB, C++, Microsoft Office Suite (Excel, Word, Powerpoint), KiCAD, Solidworks

Hardware

Soldering, Circuit Analysis, Oscilloscopes, Microcontrollers, Signal Generators, Test Equipment